



Baaz
Batch
Zindabad.

JKBOSE

🔥 2026 CLASS 11th 🔥

**SOME BASIC CONCEPT
OF CHEMISTRY**

CHEMISTRY

LECTURE-2

BY-VISHAL SIR



TOPICS TO BE COVERED

1

2

3

Classification of matter

① REVISION (Last Class)

② New lecture start

Topic teach (Chat off).

↓
End

(Chat on).

③ Lecture Notes (During lecture +
After lecture).

④ NEET level.

(Botany & Zoology) ✓

JKBOSE Wallah.

(NEET → Basics ✓)

Doubt

(Write)

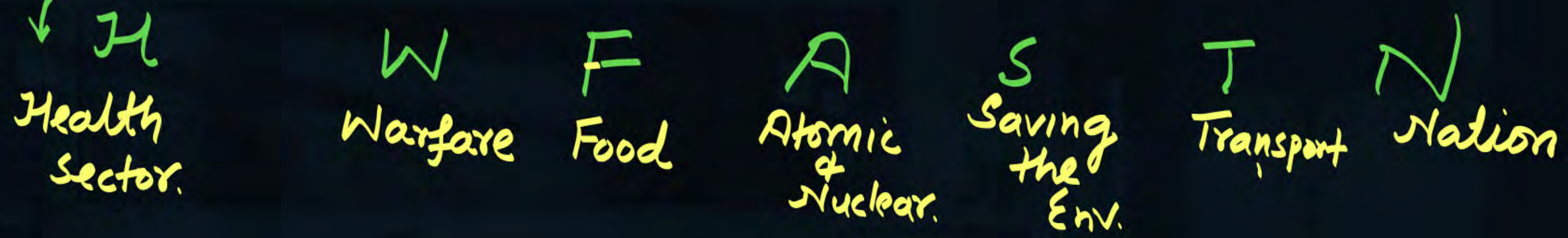
Vishal Bhaiya

Req. →

e.g. Simran Ig Id de.
Sc
No.

REVISION

Importance of Chemistry $\Rightarrow$



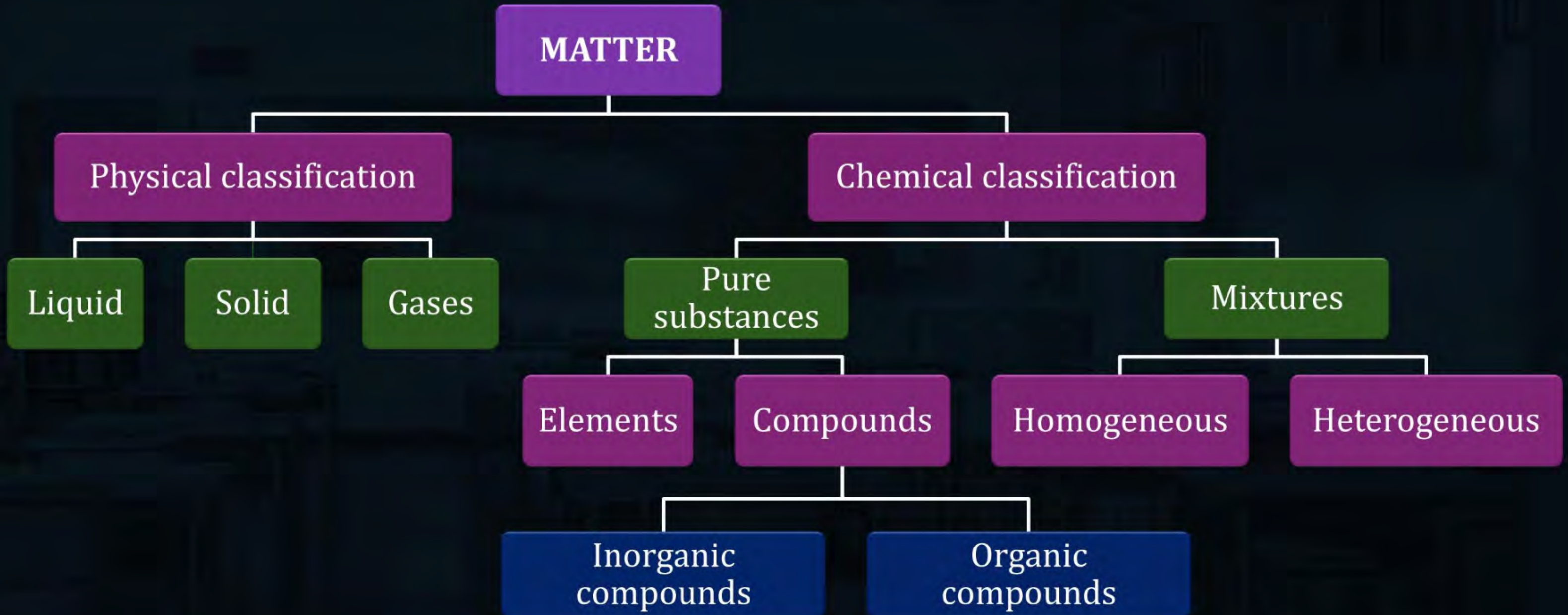
MATTER AND ITS CLASSIFICATION

- Matter is defined as any thing that occupies space[&] possesses mass.

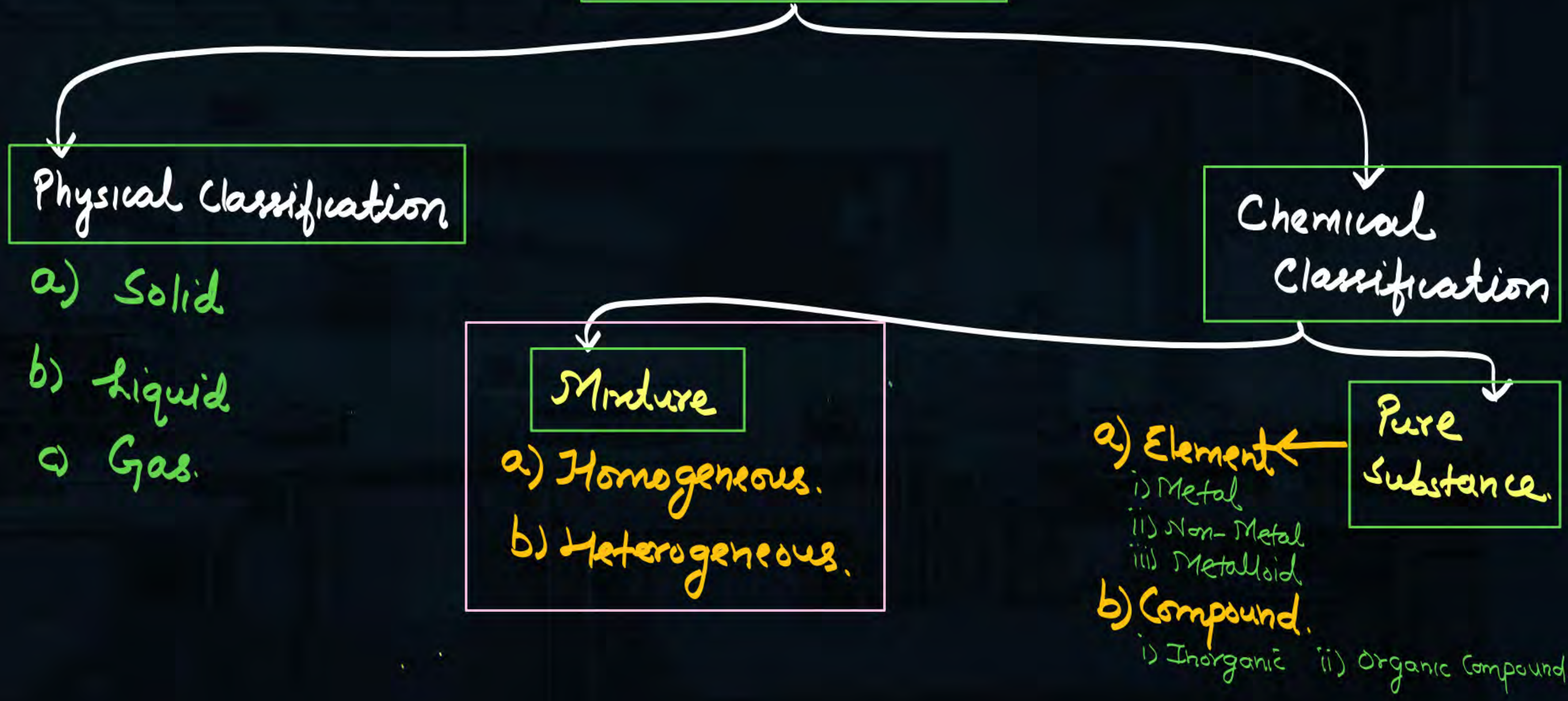
Example: A few of these include clothes, iron, gold, plastics, wood, water, milk, petrol kerosene oil, air etc.

SOUND X
Feeling X.

CLASSIFICATION



Classification of Matter.



① Physical Classification →

On the basis of physical state of matter, it is of three types.

a) Solid → A substance is said to be solid when it possess fixed shape and volume. Particles are present very close to each other & have very less movement. e.g. Pen, Pencil, Board, copy, chair etc.

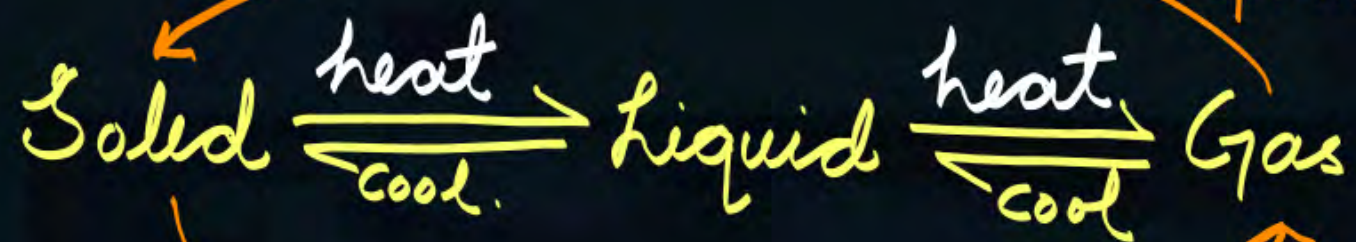
b) liquid → A substance is said to be liquid, when it possesses a fixed volume but not a fixed shape. There is little but space present between the particles and particle can move. e.g.
milk, Water, Oil, petrol, diesel etc.

c) Gaseous → A substance is said to be gaseous, when it neither possesses a fixed shaped nor a fixed volume. There is very large space present between the particles and they have enough movement. e.g. . Oxygen, Nitrogen, Carbon dioxide etc.

PHYSICAL CLASSIFICATION

All three states are interconvertible.

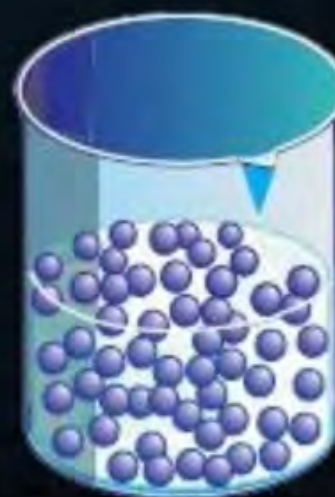
Deposition or desublimation.



Sublimation



Solid



Liquid



CHEMICAL CLASSIFICATION



2. Chemical Classification $\rightarrow$ Here matter is classified on the basis of chemical composition.

It is of two types. a) Mixture b) Pure substance.

a) Mixture $\rightarrow$ When two substances are mixed in any ratio, then it is called as mixture.

e.g. mixture of salt & sugar, sugar & water, salt & water, sand in water etc.

• Components of the mixtures can be separated by physical methods. e.g. hand picking, filtration, distillation etc.

Mixture is also of two types

i) Homogeneous mixture $\rightarrow$

Type of a mixture that have uniform composition throughout and components are not visible. e.g. Mixture of salt & water, sugar & water, air etc.

ii) Heterogeneous mixture $\rightarrow$ Type of a mixture that have non-uniform composition & components are visible. e.g. Sand in water, milk, sugar & salt etc.

CHEMICAL CLASSIFICATION



b) Pure Substance :→ Type of a matter which is made up of only one kind of particle (atom, ion, molecule).

They have fixed composition and ^{components} can not be separated by physical methods.

Pure substances are of two types.

i) Element :→ Type of pure substance that is made up of one kind of atoms is called Element.

e.g. Fe, Cu, Ag, Au, Pb, C, etc. Today we have 118 elements. Elements are also of 3 types

- Metal
- Non-Metal
- Metalloid

ii) Compound :→ Type of pure substance, that made up of two or more different elements or atoms.

e.g. H_2O , CO_2 , $CaCO_3$, H_2SO_4 , HCl etc.

Compounds are also of two types.

→ Inorganic Compounds :→ Compounds obtained from non-living things.

e.g. NaCl (Table salt), mineral, marble ($CaCO_3$), Baking soda, Plaster of Paris etc.

→ Organic Compound :→ Compounds obtained from living things e.g. Fat, carbohydrates, oil etc.

Compounds	Mixtures 8.3
1. In a compound, two or more elements are combined chemically.	1. In a mixture, two or more elements or compounds just mix together.
2. The compound contains two or more elements in a fixed ratio by mass. Its composition is always fixed.	2. The components of a mixture may be present in any ratio. Its composition is variable.
3. A compound has a definite formula.	3. A mixture does not have a definite formula.
4. A compound is always homogeneous i.e., has the same composition throughout.	4. A mixture may be homogeneous or heterogeneous.
5. A chemical reaction takes place and therefore, the formation of a compound takes place with absorption or evolution of energy.	5. No chemical reaction takes place and therefore, the formation of mixture is not accompanied by any energy change.
6. The properties of a compound are entirely different from those of its constituents.	6. A mixture shows the properties of its constituents.
7. A compound cannot be separated into its constituents by ordinary physical methods. These can be separated by chemical or electrochemical reactions.	7. A mixture can be separated into its constituents by physical methods (like filtration, evaporation, distillation, sublimation, mechanical separation etc.)
8. A compound has a fixed melting point, boiling point, etc.	8. A mixture does not have fixed melting point, boiling point, etc.

QUESTION

Which of the following methods can separate the components of a mixture?

1 Hand-picking

2 Filtration

3 Distillation

4 All of the above

Bhattacharya
on
top.

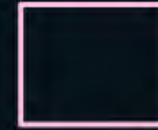
Lalke



QUESTION

Bhauchara on top

At what conditions can the states of matter interconvert?



S → L - G



1 Change in color

2 Change in temperature and pressure

3 Change in volume

4 Change in texture

QUESTION

What is the characteristic of a compound?

- 1 Made up of only one type of atom
- 2 Components can be separated by physical methods
- 3 Elements are present in a fixed ratio
- 4 Cannot undergo chemical changes



Homework

Home Work

- Identify the following substances as elements, compounds, homogeneous mixtures and heterogeneous mixtures:
- ✓ • Mercury, table salt, distilled water, air, tap water, gold, vegetable soup, petrol, milk, cloud, marble.

Lecture → Notes → Likhna



Aaj Raat ko Revise



Tab bhi Aao Rense kar

Bhauhara on
Top

Thank
You